

Multiple Choice (10 marks)

1. What is the output of "abc"[:-1] in Python 3?
 - (a) Error
 - (b) abc
 - (c) cba
 - (d) c
2. Let x = 1. What is x << 3 in Python 3?
 - (a) 1
 - (b) 3
 - (c) 8
 - (d) 16
3. Let x = 8. What is x >> 1 in Python 3?
 - (a) 3
 - (b) 4
 - (c) 2
 - (d) 8
4. In Python 3, which of the following function convert a string to an int in python?
 - (a) int(x [,base])
 - (b) long(x [,base])
 - (c) float(x)
 - (d) str(x)
5. Which of the following is correct about Python?
 - (a) It supports automatic garbage collection.
 - (b) It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.
 - (c) Both of the above.
 - (d) None of the above.

True / False (10 marks)

Answer True or False

6. True or False: $O(n)$ is larger than $O(n^2)$ when comparing their growth rates for large inputs.
7. True or False: In Python 3, the output of `print(list[1:3])` with the given list is `[786, 2.23]`.
8. True or False: In Python 3, using `b[::2]` on the list `[11, 13, 15, 17, 19, 21]` results in the output `[11, 15, 19]`.
9. True or False: A group of cookies stored by the user's web browser can significantly compromise a user's personal privacy.
10. True or False: In Python, applying the lambda function `r` to the argument 3 results in the value 6.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. After extensively testing a large Java program with no errors found, discuss the implications of this outcome and explain why it's still possible for the program to contain bugs.
12. Discuss the purpose of the `'isupper()'` function in Python 3. How does it determine if all characters in a string are uppercase, and what are potential use cases for this function?

End of Paper